

CMPG 325 COMPUTER NETWORKS

Individual Semester Project Milestone 1: Client Design Review

1. Client Requirements

The Greater Taung Local Municipality Offices require a reliable computer network to support communication and the sharing of resources between different municipal departments.

The main client requirements identified are:

- Provide reliable network connectivity between the different municipal offices and departments.
- Allow staff computers to communicate with each other and access shared network resources.
- Provide access to important services such as file sharing, printing and municipal applications.
- Provide a server to support the required network services.
- Provide guest Wi-Fi for visitors and isolate guest users from internal network resources.
- Ensure that critical file, print and application services remain available during business hours.
- Configure Basic OSPF (single-area dynamic routing) between the network sections.
- Produce a working and testable Cisco Packet Tracer implementation using the assigned addressing block of 172.30.70.0/23.

2. Physical Topology

The physical topology shows the physical arrangement of the network devices and how they are connected. The topology includes the routers, switches, computers, server and wireless access point required by the municipality.

Main Network Devices:

- **Core-Router (R1):** The Core-Router provides routing for the internal network and is connected to R2 through a point-to-point network link.
- **Branch Router(R2):** R2 provides connectivity between the router link and the VLAN-based network sections. It also uses router sub interfaces to provide the gateways for VLAN 10 and VLAN 20.
- **Internal Switch (SW1):** The Internal Switch provides wired connectivity for the internal municipal devices connected to the Core-Router side of the network.
- **Access Switch (SW2):** The Access Switch provides connectivity for the VLAN-based devices. It is used to connect the internal and guest network devices to the appropriate VLANs.
- **Municipal-server:** The Municipal-server represents the server used to provide important network services such as file, print and application services.
- **Admin-PC:** Represents an administrative workstation on the internal network.
- **Finance-PC:** Represents a finance workstation on the internal network.
- **HR-PC:** Represents an internal municipal workstation connected through VLAN 10.
- **Services-PC:** Represents an additional internal workstation connected through VLAN 10.

- **Guest-AP (Wireless Access Point):** Provides wireless connectivity for guest users.
- **Guest-Client:** Represents a visitor device that connects through the guest network.

The physical topology diagram was created using draw.io and shows the physical connections between the network devices.

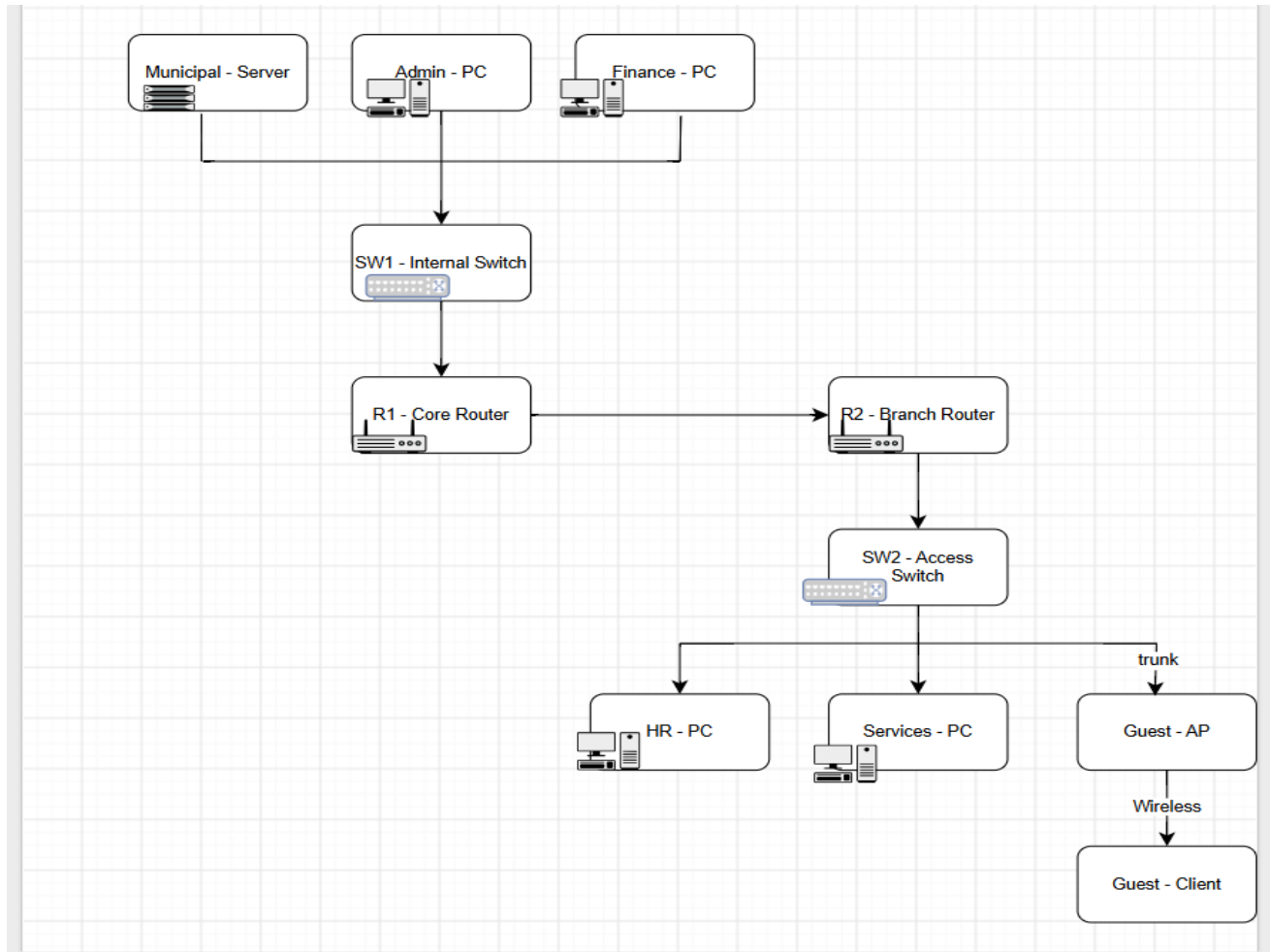


Figure 1: Physical Topology (draw.io)

3. Logical Topology

The logical topology was designed and implemented in Cisco Packet Tracer. It shows how the network is logically organised, including the routers, switches, VLANs and network connections.

The network uses two VLANs:

- **VLAN 10 - INTERNAL:** Used for internal municipal users and devices. The HR-PC and Services-PC are assigned to this VLAN.
- **VLAN 20 - GUEST:** Used for guest users and devices. The Guest-Client is connected to the guest network through the wireless access point (Guest-AP).

Routing and Guest Network Isolation

The assigned networking challenge for the project is Basic OSPF (single-area dynamic routing). OSPF is configured on the routers so that routing information can be exchanged dynamically between the network sections.

This allows the routers to learn the available networks without requiring every route to be manually entered. The OSPF configuration is also part of the testing and verification that will be demonstrated as part of the project.

Guest VLAN 20 is excluded from the OSPF network statements configured on R2, so the Core-Router never learns a route to the guest subnet. This enforces the isolation required by CR3, ensuring that guest devices cannot reach internal municipal resources while still allowing guest users to access the wireless network.

The logical topology was implemented in Cisco Packet Tracer and is used to simulate the operation of the municipal network.

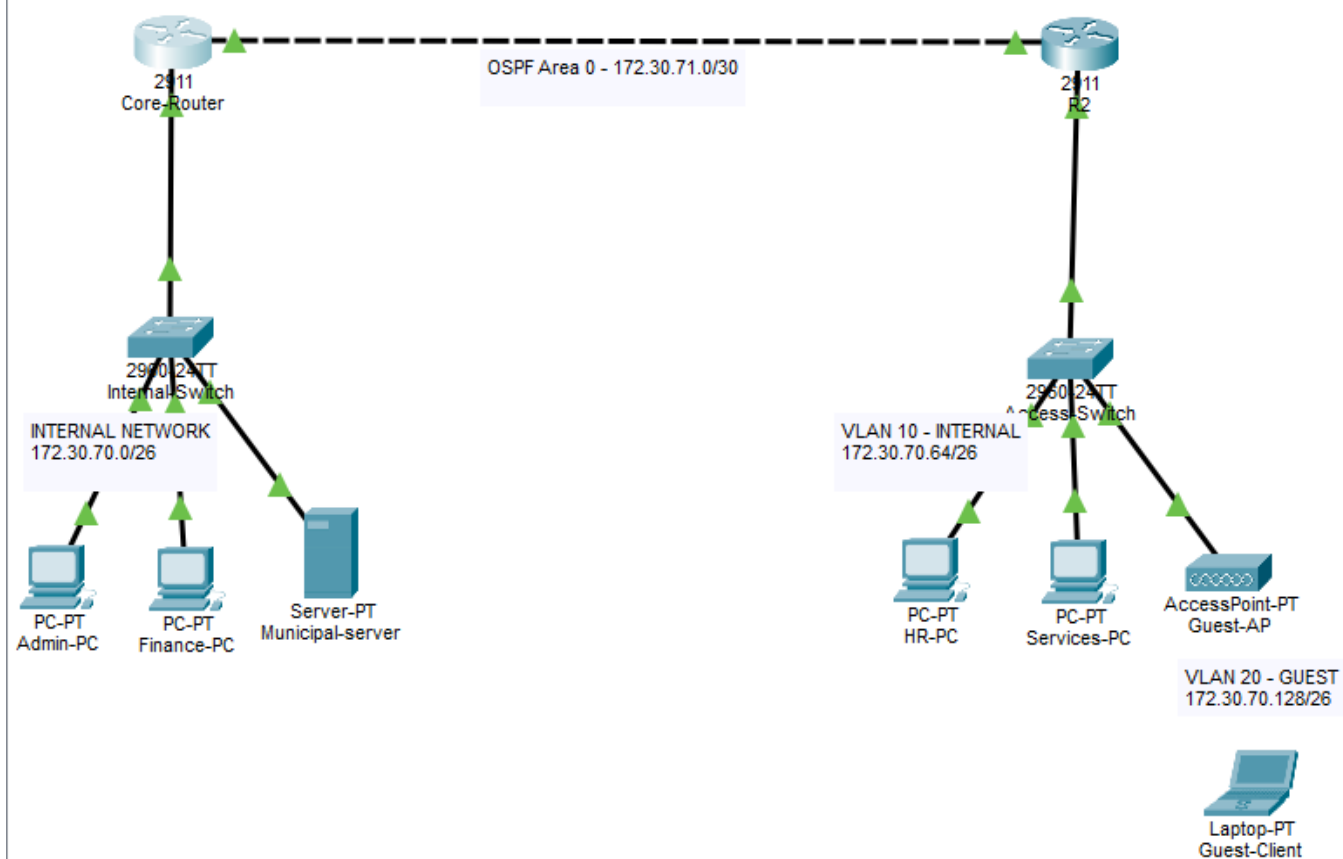


Figure 2: Logical Topology (Cisco Packet Tracer)

4. IP Addressing Plan

The network uses the private address range 172.30.70.0/23. The address range is divided into smaller subnets to accommodate the different network segments.

Network Segment	Network Address	Subnet Mask	Default Gateway
Core-Router Internal LAN	172.30.70.0/26	255.255.255.192	172.30.70.1
Internal VLAN 10	172.30.70.64/26	255.255.255.192	172.30.70.65
Guest VLAN 20	172.30.70.128/26	255.255.255.192	172.30.70.129
Core-Router-R2 Link	172.30.71.0/30	255.255.255.252	—

The router interfaces are configured as follows:

Device	Interface	IP Address
Core-Router (R1)	GigabitEthernet0/1	172.30.70.1
Core-Router (R1)	GigabitEthernet0/0	172.30.71.1
R2	GigabitEthernet0/0	172.30.71.2
R2	GigabitEthernet0/1.10	172.30.70.65
R2	GigabitEthernet0/1.20	172.30.70.129

Individual devices on the network are addressed as follows:

Device	IP Address	Subnet Mask	Default Gateway
Admin-PC	172.30.70.11	255.255.255.192	172.30.70.1
Finance-PC	172.30.70.12	255.255.255.192	172.30.70.1
Municipal-Server	172.30.70.10	255.255.255.192	172.30.70.1
HR-PC (VLAN 10)	172.30.70.66	255.255.255.192	172.30.70.65
Services-PC (VLAN 10)	172.30.70.67	255.255.255.192	172.30.70.65
Guest-Client (VLAN 20)	172.30.70.130	255.255.255.192	172.30.70.129

R1's internal LAN uses the 172.30.70.0/26 subnet, VLAN 10 uses the 172.30.70.64/26 subnet, and VLAN 20 uses the 172.30.70.128/26 subnet. The remaining addresses in the original /23 network can be used for future network expansion.

5. Initial GitHub Repository

<https://github.com/Phathu647/CMPG325-Greater-Taung-Municipal-Network.git>